

Lower Limb Auxiliary Exoskeleton with Gait Management Feedback

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Grade 10

Why?

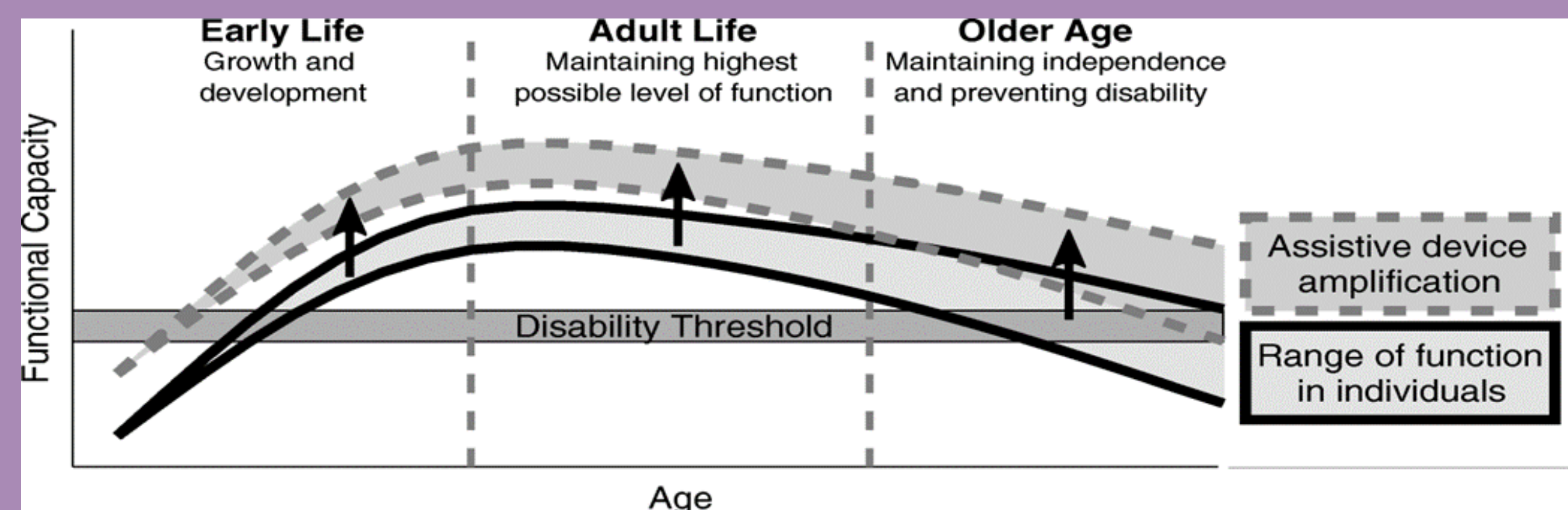
Flaws within traditional mobility aids such as canes and scooters:

1. Require strong upper body strength to operate
2. Do not address the complex nuances of gait and muscle engagement
3. Can cause hunching (fig. 1)



These flaws can be extremely detrimental to the two main groups of people requiring mobility aids:

1. Seniors (fig. 2)
2. Individuals with incorrect gait



Exoskeleton technology can counter the flaws because it can:

1. Provide more structural support to the lower body
2. Be more customizable than conventional aids in that different sensors and electronic modules can be added to help with gait management

ENGINEERING GOAL

Design a mobility aid to address the flaws of traditional mobility aids and *provide lower body support with gait management* by controlling 4 Servos and using pressure sensors to create a *feedback loop for gait analysis*.

Design Overview

Mechanical Design

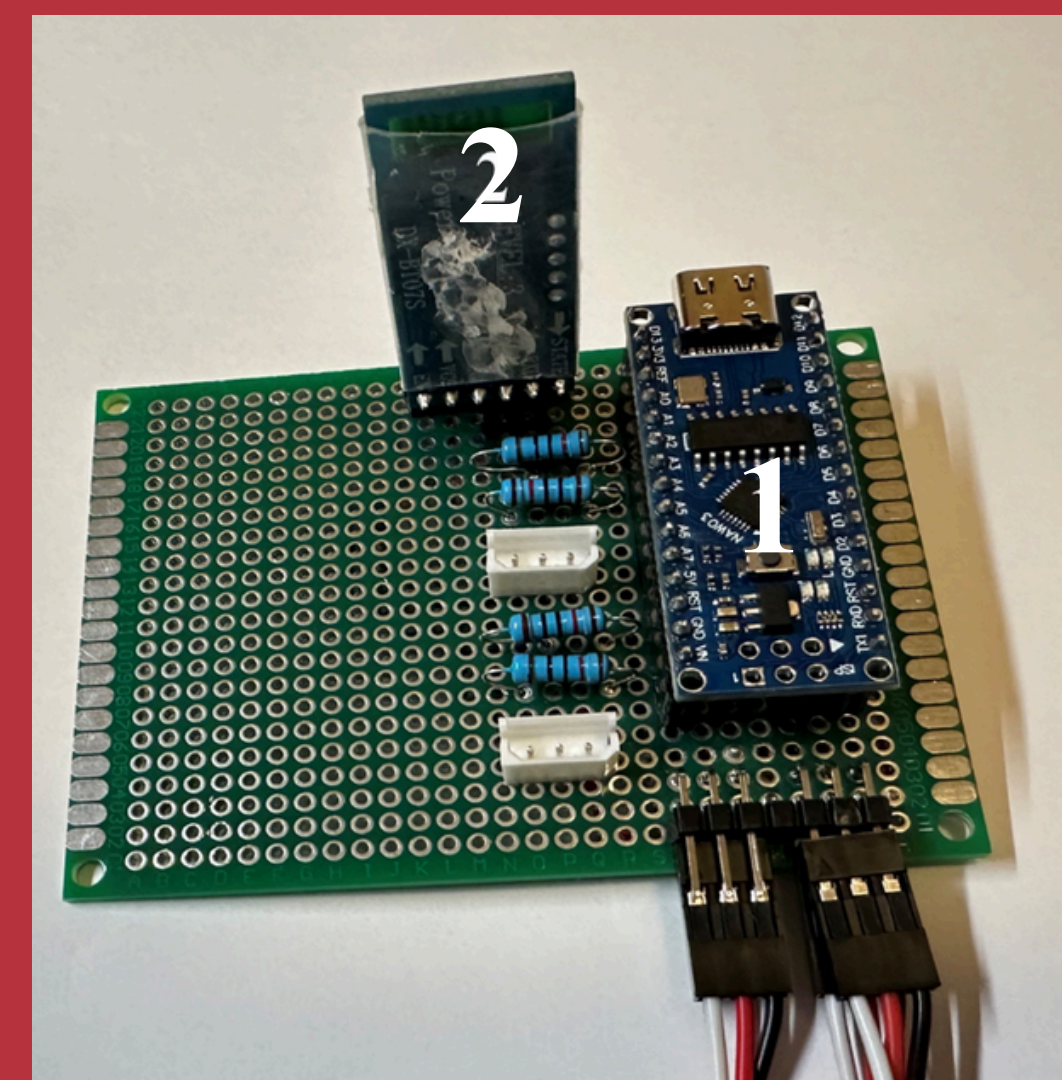
1. Main Skeletal Framework
2. DH-03X Large Torque Alloy Servos
 - a. Powered by a 12V rechargeable battery that can run for up to 4 continuous hours
 - b. 0.595 kg per Servo
 - c. Torque: 180 kg.cm (≈ 17.65 Nm)
 - d. $\approx \$55.34$ per Servo
3. 3D Printed and Laser Cut Components
 - a. Two pairs of C-shaped, 3D-printed elements that are securely affixed to each metal bar
 - b. System of three laser-cut acrylic components designed to govern the movements of the calf section
4. Torso Attachment
 - a. Encircles the user's waist to provide support for the spinal cord
 - b. Designed to help with the user's posture



Electronics Design

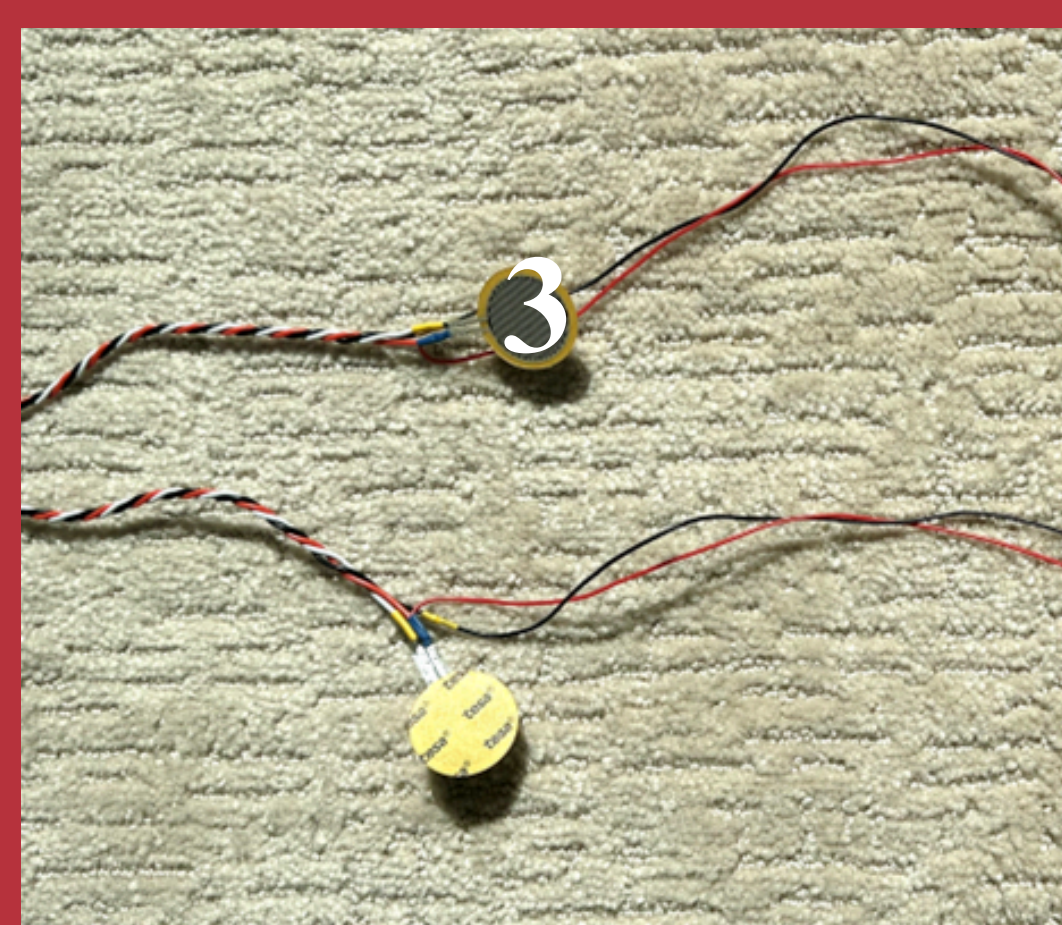
Electric Components

1. Arduino Nano Board: the central control module
2. Bluetooth module: for easy control
The Bluetooth Blinker app allows the users to start and stop the exoskeleton and switch between gait analysis and gait correction modes by clicking different buttons.
3. Pressure sensors: for gait analysis



Code Design

1. "map ()" Function
 - a. `map (a, 0, 1023, 0, 255);`
 - b. `ADC  $\rightarrow$  PWM  $\rightarrow$  0°-180°`
2. Bluetooth Module
 - a. Uses the library "Blinker.h" and app "Blinker"
 - b. Incorporates Boolean variable to dictate the start (false) and stop (true) of the exoskeleton's movements



Gait Data Collection Mode

I used a data collection device with potentiometers to record the rotational degrees of my hip and knee movements when I walk.

- The device is made of thin wood planks and four potentiometers placed at the joints

After the potentiometers relay the information to the Arduino Nano board, the numbers are printed in Arduino’s Serial Monitor.

A plug-in written by Indrek Luuk called ArduSpreadsheet can export the data into an Excel file

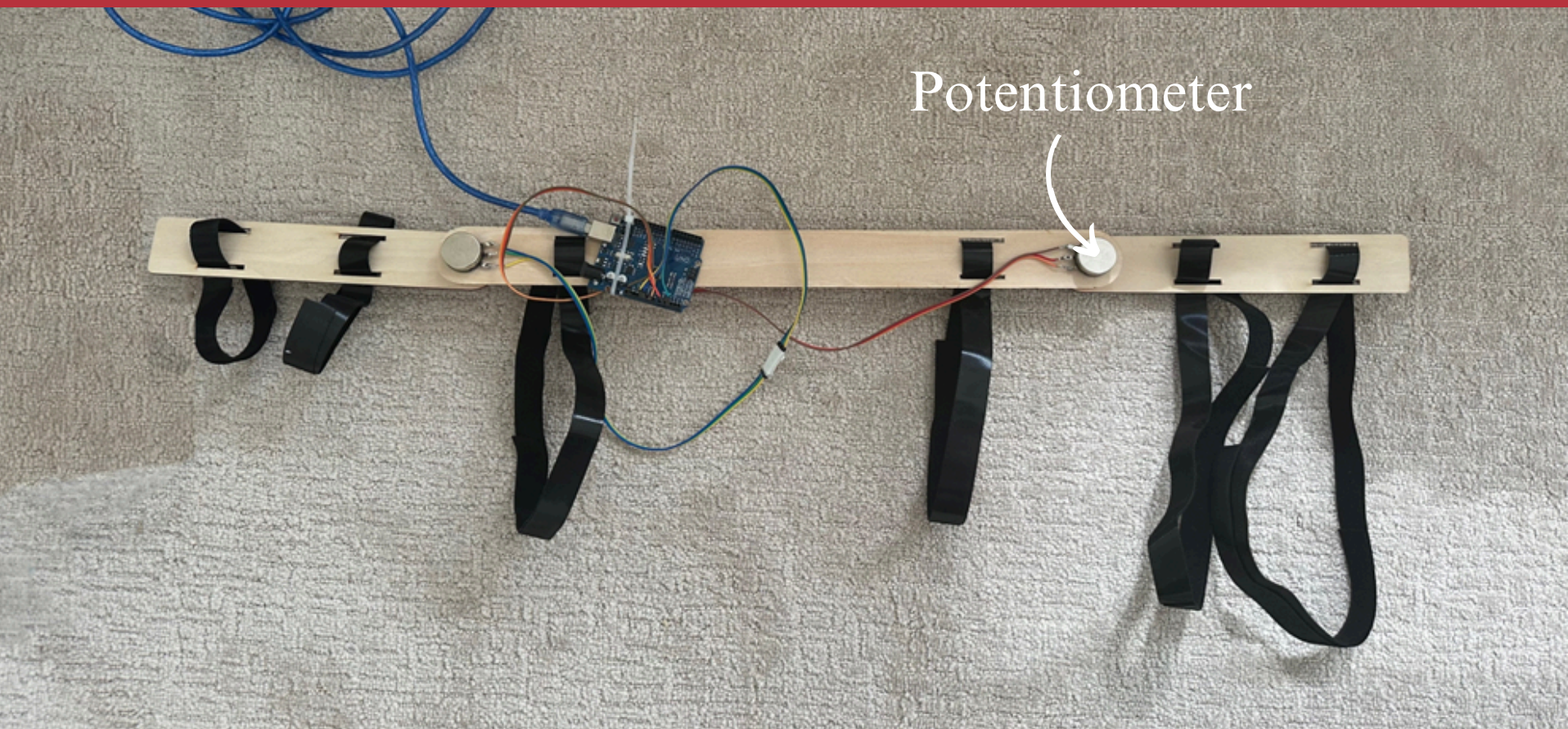


Image of the data collection device

05:14.5	-7	21
05:14.5	-7	20
05:14.5	-7	20
05:14.5	-7	20
05:14.5	-6	20
05:14.6	-6	20
05:14.6	-6	20
05:14.6	-5	20
05:14.6	-5	20
05:14.6	-5	19
05:14.6	-5	19
05:14.6	-5	19
05:14.6	-5	19
05:14.6	-4	19

Example of data exported from ArduSpreadsheet

Left: Time stamps

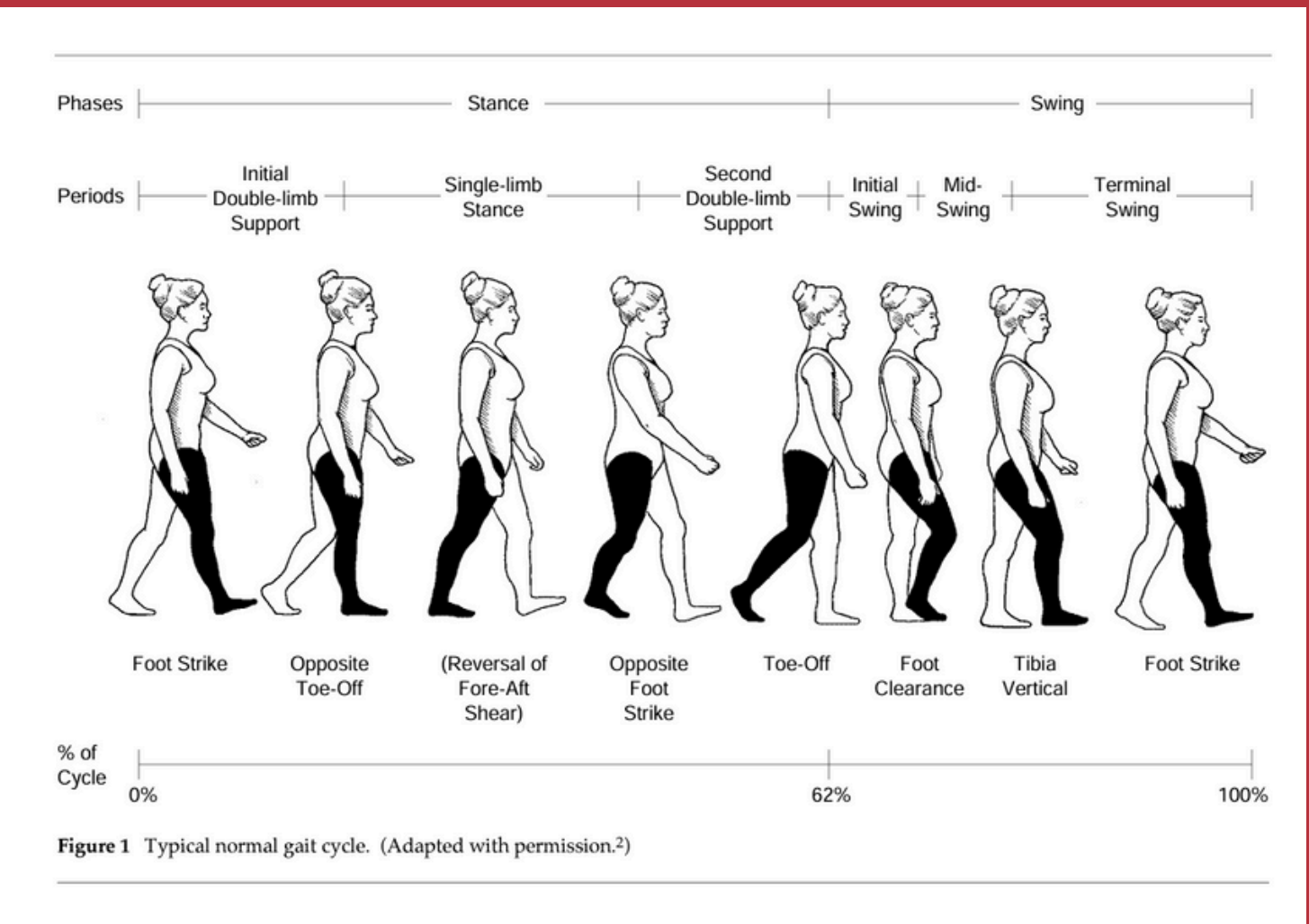
Middle: Degrees of Right Hip Movement

Right: Degrees of Right Knee Movement

Gait Correction Mode

The large torque of the Servos forces the user to walk with a healthy gait.

Calculated joint angles using inverse kinematics
Modified constants to control the rotational speed and step distance
according to the typical normal gait cycle



Typical Normal Gait Cycle (Chambers et al. 2002)

Results Discussion

This exoskeleton is successful in reaching my engineering goal because it can:

- 1.Start, stop, and switch modes on the user's command through Bluetooth
2. Provide support to lower body
3. Show changes in pressure values when the user walks in different gaits

Gait Analysis Through Feedback Loop

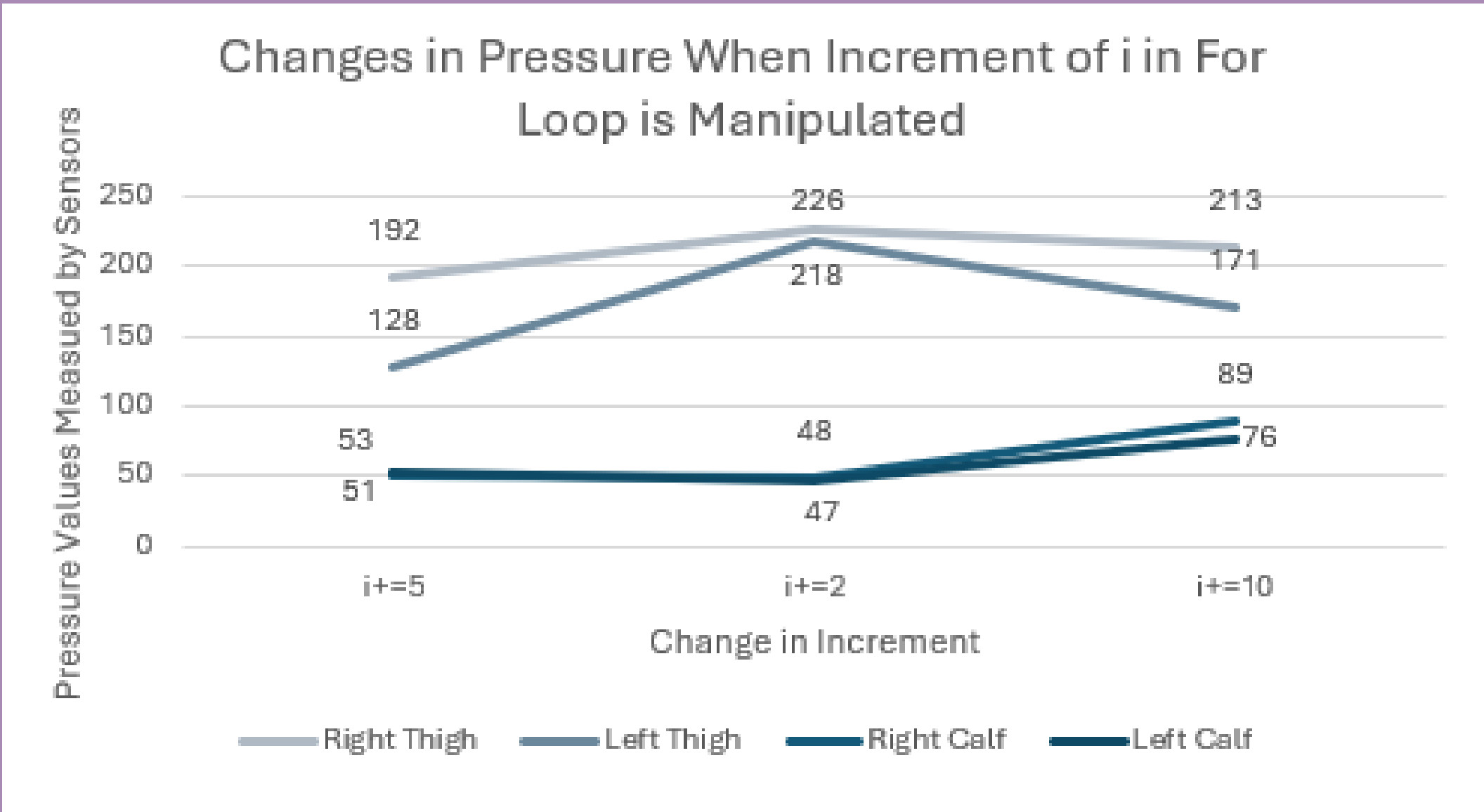
	Left Quad	Left Calf	Right Quad	Right Calf
Normal Gait	128	53	192	51
Calf Pushes Off Ground Hard	210	69	218	78
Quad Pushes Against Exoskeleton	317	48	356	53

Pressure values are in ADC in both graphs

Lower Body Support

- It does not require upper body strength
- Torso attachment provides structural support for the spinal cord and prevents hunching
- Rigid torque prevents the user's knees from buckling and thus prevents the user from falling

Gait Analysis Through Changing Constants



There were significant changes in pressure values after i was changed, suggesting that speed manipulation is another way of providing data for gait analysis and that my feedback loop was successful in relaying back changes in gait

So What?

Gait Analysis Through Feedback Loop Discussion

- Gait when calf pushes off the ground hard
 - This gait is the most similar to walking patterns caused by equinus. However, equinus is primarily associated with neurological factors such as cerebral palsy, so there should still be significant differences between a true equinus gait and mine (Physiopedia).
- Gait when the thighs push against the exoskeleton hard
 - This gait is the most similar to the posterior lurching gait, where the hip is extended and the individual lacks glute engagement (Ataullah et al., 2023). I felt my glutes relaxed substantially when I tensed my quads to push against the exoskeleton’s leg attachments, but the results I have gathered from my sensors do not indicate this relaxation.

What’s Next

The pressure sensors should be replaced with sensors such as muscle sensors (electromyography) that are more accurate when testing for muscle engagement and activity

- More sensors can be added to other major muscle regions, such as the glutes
- Infrared or ultrasonic distance-measuring sensors can detect the height of terrain

Use lighter and more versatile Servos such as the AK80-9 48V Robotic Dynamic which weighs 0.485 kg and has a peak torque of 18 Nm.

- Motor has to be back drivable
 - Allows users to move their legs freely with little resistance to perform daily tasks such as standing up and climbing stairs

Implement fall detection and add safety measures

Thanks and References

- I would like to thank my coding teacher, Sunny, for taking the time to look over my code whenever I needed help.
- I would also like to thank my mother for supporting me through this entire project and purchasing whatever materials I needed, even if I made several purchase mistakes.

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• Software

- AutoCAD
- Autodesk Fusion360
- Arduino (software and hardware)
- Visual Studio Code